INSTITUTO SUPERIOR DE AGRONOMIA
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CLIMATE VARIABILITY AND YIELD STABILITY OF MAJOR STAPLE CROPS IN SUB-SAHARAN AFRICA

ANALYSIS AND VISUALISATION OF COMPLEX AGRO-ENVIRONMENTAL DATA FINAL PROJECT

Master's in Green Data Science

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Table of Contents

1. Introduction

Climate variability is widely recognised as one of the principal sources of yield risk for rainfed staple crop production in Sub-Saharan Africa (SSA), where the great majority of smallholder farming systems remain dependent on seasonal rainfall and have limited access to irrigation. Recent decades have seen growing evidence of shifting rainfall patterns and rising temperatures across the region, alongside an apparent increase in the frequency of extreme rainfall events and associated flooding, most visibly illustrated by the catastrophic 2022 Nigeria floods, which displaced over 1.4 million people and damaged more than 332,000 hectares of farmland (Wikipedia, 2024; World Weather Attribution, 2022). Understanding whether these climate shifts are statistically detectable over the long run, and how they translate into yield-level and yield-stability outcomes across different staple crops and countries, is essential for anticipating future food security risk and for guiding adaptation priorities in the region.

This project examines four decades (1984–2024) of national-level yield and climate data for three staple crops, maize, millet and cassava, across five Sub-Saharan African countries: Nigeria, Ghana, Kenya, Mali and Ethiopia. These three crops were deliberately selected to span a gradient of expected climate sensitivity, from the relatively water-demanding maize, to the more drought-tolerant millet, to cassava, a root crop generally considered more climate-resilient by virtue of its physiology and deep rooting structure. The five countries similarly span distinct climatic zones, from the Sahel-adjacent north of Nigeria and Mali, to the more humid coastal zone of Ghana, to the East African highland and Rift Valley climates of Kenya and Ethiopia, allowing comparison of climate vulnerability across contrasting agroecological settings.

The objective of this study is to determine whether long-term rainfall and temperature patterns in these five countries have changed significantly over the study period, to quantify the relationship between climate variability and crop yield, and to assess whether yield stability and climate vulnerability differ systematically by crop and by country.

1.1 Research Questions and Hypotheses

Question 1 (Climate Trend): Have temperature and rainfall patterns in the study countries changed significantly over the 1984–2024 period?

$H0_1$: There have been no significant changes in temperature and rainfall patterns over the study period.

Question 2 (Climate–Yield Relationship): Do climate variables (rainfall, temperature, solar radiation) have a significant effect on crop yield?

$H0_2$: Climate variables have no significant effect on crop yields.

Question 3 (Yield Stability): Does yield stability differ significantly among staple crops?

$H0_3$: There are no significant differences in yield stability among staple crops.

Question 4 (Regional Differences): Does climate vulnerability differ significantly among countries/regions?

$H0_4$: There are no differences in climate vulnerability among African regions.

2. Database Description

The dataset used in this project merges two independently sourced data series. Crop yield, area harvested and production data for Maize, Millet and Cassava were obtained from FAOSTAT (Crops and Livestock Products domain, FAO, 2024) for Ethiopia, Ghana, Kenya, Mali and Nigeria. Climate data, rainfall, mean temperature and solar radiation, were obtained from NASA's POWER (Prediction of Worldwide Energy Resources) Agroclimatology archive (NASA POWER, 2024), following these steps:

i. FAOSTAT data was downloaded in long format (2,496 rows × 15 columns) covering three elements (Area harvested, Yield, Production), three crops, five countries, and years 1961–2024.

ii. Data was audited for quality: no missing values were found; a minority of records were flagged as estimated or imputed by FAOSTAT, which is standard practice and were retained.

iii. The long-format table was reshaped (pivoted) to wide format (one row per Country × Crop × Year) and yield was converted from FAOSTAT's native kg/ha to t/ha for interpretability.

iv. One representative agricultural coordinate per country was selected (e.g. Guinea Savanna belt for Nigeria; Rift Valley for Kenya) and monthly climate data were pulled from the NASA POWER API for three parameters: precipitation, mean temperature, and solar radiation.

v. Monthly climate values were aggregated to annual totals/means and saved as a country-year climate file. The start year was capped at 1984 (the earliest year all three NASA POWER parameters are jointly available).

vi. The cleaned FAOSTAT yield file and the NASA POWER climate file were merged on Country and Year (inner join), producing the final analytical dataset of 556 observations × 9 columns, covering 1984–2024

Table 1. Scope of the merged dataset by crop and country.

Crop	Ethiopia	Ghana	Kenya	Mali	Nigeria
Maize	32	41	41	41	41
Millet	32	41	41	41	41
Cassava	0	41	41	41	41

Data coverage is unbalanced in two respects, both substantive rather than incidental. Ethiopia's series begins in 1993, 11 years later than the other four countries, because Ethiopia's national statistical system was reorganised following the 1991 political transition. Ethiopia also has no cassava production recorded, consistent with the crop's limited agroecological suitability and cultural relevance in the Ethiopian highlands relative to West African contexts; cassava-specific analyses are therefore conducted across the four remaining countries only.

Key variables

Three outcome variables are recorded for every crop-country-year observation: area harvested (hectares), production quantity (tonnes), and yield (tonnes per hectare, converted from FAOSTAT's native hg/ha for interpretability). Three climate predictor variables are recorded for every country-year: total annual rainfall (mm/year, aggregated from monthly NASA POWER values), mean annual temperature (°C), and mean daily solar radiation (MJ/m²/day). There are no missing values in any of the nine columns across the 556 merged observations.

Table 2. Descriptive statistics for grain yield (t/ha), by crop.

Crop	n	Mean	SD	Min	Max
Maize	196	1.81	0.57	0.96	3.71
Millet	196	0.99	0.45	0.27	2.56
Cassava	164	11.77	4.13	5.00	25.86

Table 3. Descriptive statistics for climate variables

Country	Years	Mean Rainfall (mm/yr)	Mean Temp (°C)	Mean Solar Rad. (MJ/m²/day)
Nigeria	1984–2024	1759.6	23.4	20.7
Ghana	1984–2024	1109.2	27.0	19.2
Kenya	1984–2024	1919.8	18.7	21.3
Mali	1984–2024	1007.6	28.4	21.0
Ethiopia	1993–2024	1808.7	20.0	20.9

Data quality notes

Two data quality considerations are documented here for transparency. First, FAOSTAT flags a small minority of records as “estimated” rather than “official figure”, consistent with normal FAOSTAT practice for years or countries where official government reporting is incomplete; these are retained, as exclusion would introduce additional gaps rather than improve accuracy. Second, Nigeria's 2020–2022 rainfall values are markedly elevated relative to its long-run average; this was cross-checked against independent flood-attribution literature and is corroborated as a genuine climate anomaly rather than a sensor or model artefact, and is therefore retained in all analyses.

3. Exploratory Data Analysis

3.1 Climate Trend Analysis (Hypothesis 1)

A Mann-Kendall trend test was applied to the annual rainfall and mean temperature series of each country to test for a statistically significant monotonic trend over the study period (Table 4 and Figure 1).

Table 4. Mann-Kendall trend test results for rainfall and temperature, by country.

Country	Rainfall Trend	Rainfall p-value	Temp. Trend	Temp. p-value	Temp. Slope (°C/yr)
Nigeria	no trend	0.814	increasing	0.003 **	0.0372
Ghana	no trend	0.108	increasing	0.006 **	0.0204
Kenya	increasing	0.036 *	no trend	0.185	0.0061
Mali	increasing	0.004 **	no trend	0.340	0.0065
Ethiopia	no trend	0.092	increasing	0.026 *	0.0188

** p < 0.05, ** p < 0.01. Slope is the Sen's slope estimate from the Mann-Kendall test.*

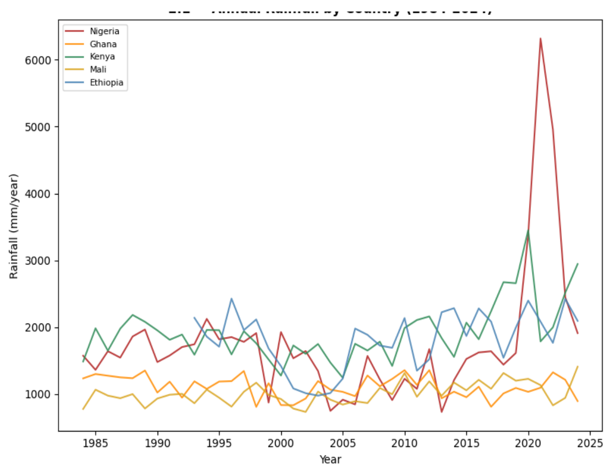

Figure 1a: Annual Rainfall by country

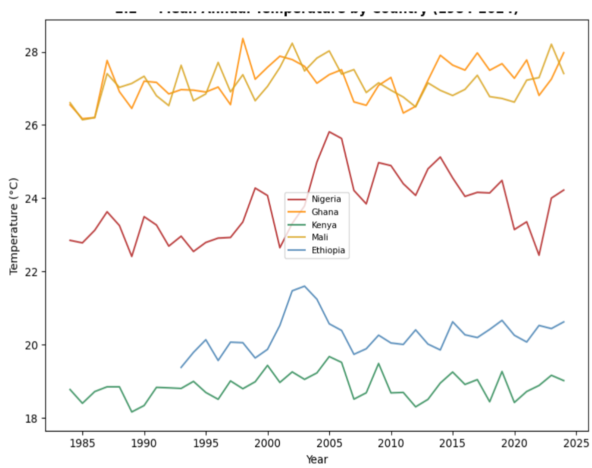

Figure 1b: Annual Temperature by country

Temperature shows a statistically significant warming trend in three of the five countries: Nigeria ($p = 0.003$, +0.037 °C/year), Ghana ($p = 0.006$, +0.020 °C/year) and Ethiopia ($p = 0.026$, +0.019 °C/year). Rainfall shows a statistically significant increasing trend in two countries: Kenya ($p = 0.036$) and Mali ($p = 0.004$). Across the ten country-variable combinations tested, five show a statistically significant trend ($p < 0.05$). $H0_1$ is therefore rejected for these five combinations, and the overall evidence supports $H1_1$: temperature and rainfall patterns have changed significantly over the study period, though the result is heterogeneous rather than uniform across countries and variables, with no country showing significant trends in both rainfall and temperature simultaneously.

3.2 Climate–Yield Relationship (Hypothesis 2)

Correlations between yield and the three climate variables differ in both magnitude and direction across crops (Table 5). Rainfall is positively correlated with yield in Maize ($r = 0.177$) and Millet ($r = 0.152$), but negatively correlated in Cassava ($r = -0.231$). Temperature is negatively correlated with yield in Maize and Millet, but positively correlated in Cassava ($r = 0.299$). Solar radiation is negatively correlated with yield in all three crops, most strongly in Cassava ($r = -0.331$).

Table 5. Correlation (r) between yield and climate variables, by crop.

Crop	Yield vs Rainfall	Yield vs Temp.	Yield vs Solar Rad.
Maize	0.177	-0.083	-0.064
Millet	0.152	-0.057	-0.166
Cassava	-0.231	0.299	-0.331

Multiple linear regression of yield on all three climate variables was significant overall for all three crops (Table 6), confirming that climate variables jointly explain a statistically significant share of yield variance in every crop tested, though the proportion of variance explained (R^2) is modest throughout, ranging from 4.5% in Maize to 13.5% in Cassava.

Table 6. Multiple linear regression of yield on climate variables, by crop.

Crop	R^2	Adj. R^2	F-statistic	p(F)	n
Maize	0.045	0.030	3.040	0.030 *	196
Millet	0.077	0.063	5.358	0.002 **	196
Cassava	0.135	0.119	8.338	<0.001 ***	164

** $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. $H0_2$ is rejected for all three crops: climate variables jointly have a statistically significant effect on yield.*

At the individual-predictor level, solar radiation is the most consistently significant climate variable, with a significant negative coefficient in Millet ($p = 0.001$) and Cassava ($p = 0.007$). Rainfall is only marginally significant in Maize ($p = 0.063$) and not significant in Millet or Cassava at conventional thresholds in the multiple regression, despite the simple correlations reported above; this divergence reflects the substantial multicollinearity among the three climate predictors (Table 7), which inflates coefficient standard errors and makes individual-predictor significance tests unreliable even where the overall model is significant.

Table 7. Variance Inflation Factor (VIF) diagnostics, by crop.

Crop	Rainfall VIF	Temp. VIF	Solar Rad. VIF
Maize	8.3	42.4	65.5
Millet	8.3	42.4	65.5
Cassava	7.3	42.9	63.4

VIF values above 10 are conventionally considered indicative of problematic multicollinearity; temperature and solar radiation both far exceed this threshold in all three crops, reflecting that these two variables are themselves strongly correlated at the country

Because of this multicollinearity, a standardised single-predictor sensitivity analysis (regressing yield separately on each standardised climate variable) was used to assess relative climate sensitivity more reliably (Table 8). This analysis shows a markedly different sensitivity profile for Cassava relative to the two cereals: rainfall, temperature and solar radiation are all statistically significant predictors of Cassava yield ($p < 0.01$ for all three), with comparatively large standardised slopes, whereas Maize and Millet show smaller, more selective sensitivity.

Table 8. Standardised single-predictor sensitivity of yield to each climate variable, by crop.

Crop	Variable	Standardised Slope	p-value
Maize	Rainfall	0.101	0.013 *
Maize	Temperature	-0.048	0.246
Maize	Solar Radiation	-0.037	0.374
Millet	Rainfall	0.068	0.033 *
Millet	Temperature	-0.026	0.428
Millet	Solar Radiation	-0.075	0.020 *
Cassava	Rainfall	-0.949	0.003 **
Cassava	Temperature	1.230	<0.001 ***
Cassava	Solar Radiation	-1.364	<0.001 ***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

3.3 Yield Stability Across Crops (Hypothesis 3)

Figure 2 shows the yield trends by country and crop. Yield coefficient of variation (CV%), computed per country-crop combination over the full study period, was used as the measure of yield stability (Table 9). CV% ranges from 10.6% (Maize, Kenya) to 43.0% (Millet, Ethiopia), with substantial overlap in the distributions across all three crops.

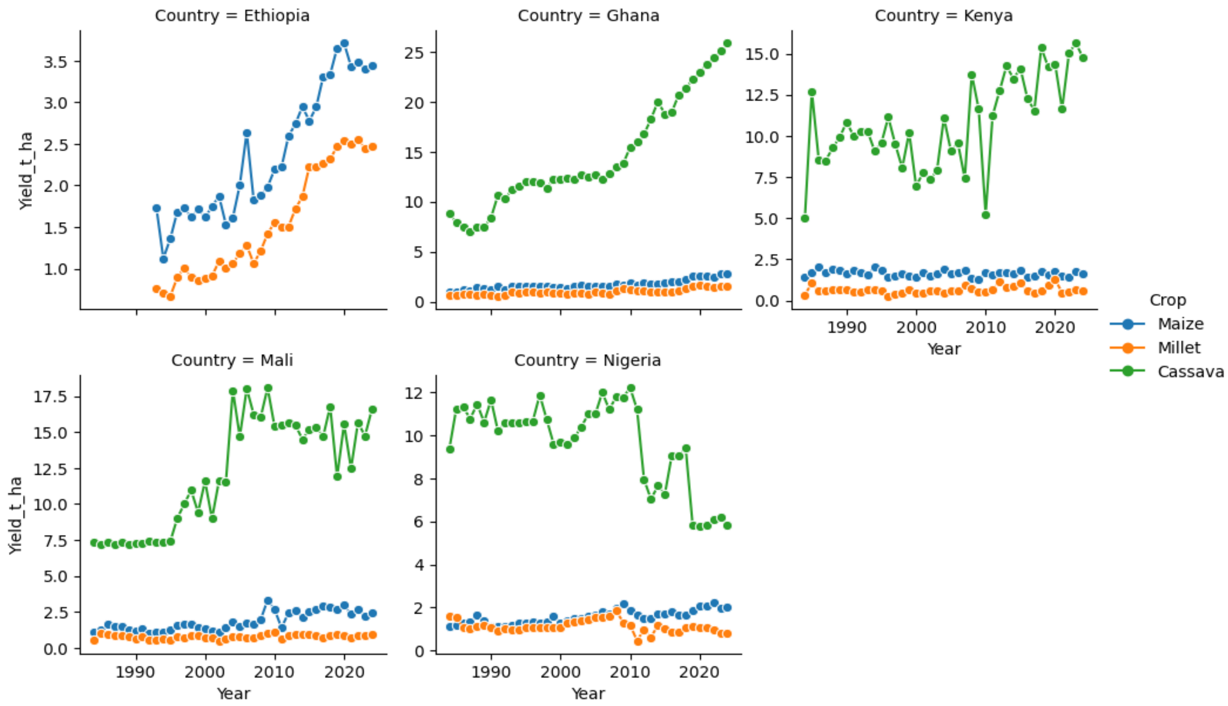

Figure 2a: Yield trends by country and crop

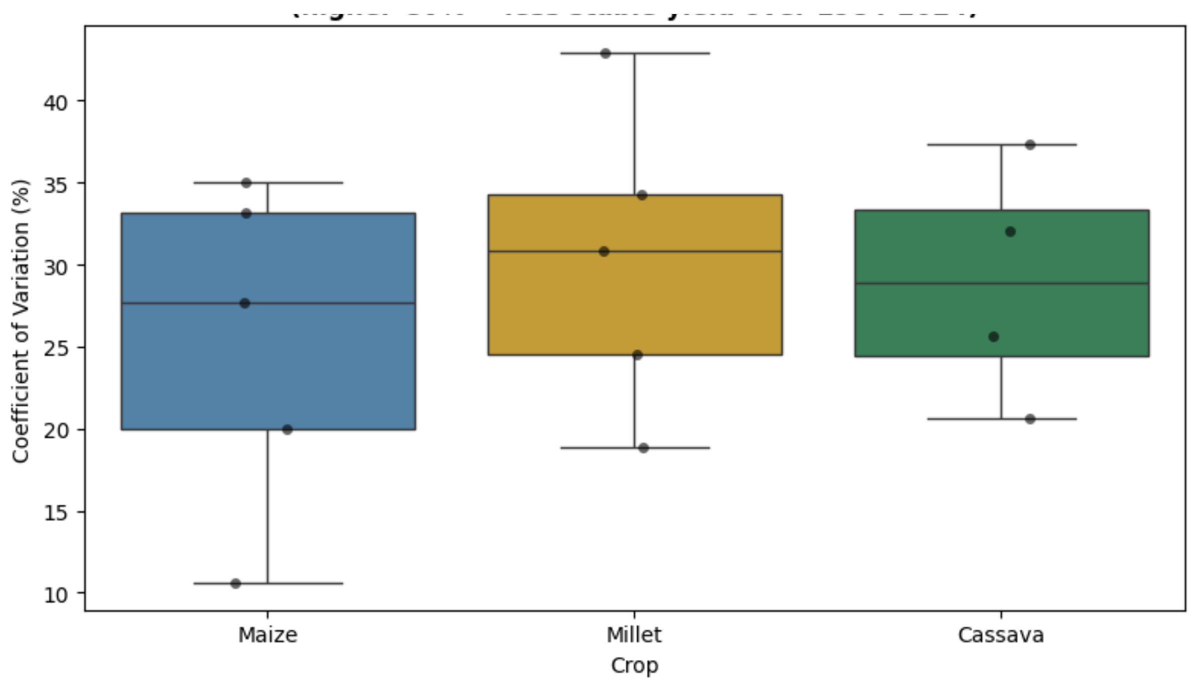

Figure 2b: Yield variability (CV%) by crop, across countries (1984 – 2024)

Table 9. Yield coefficient of variation (CV%) by country and crop.

Country	Crop	Mean Yield (t/ha)	CV (%)
Kenya	Maize	1.66	10.56
Nigeria	Cassava	9.68	20.64
Nigeria	Maize	1.57	20.00
Mali	Millet	0.80	18.86
Nigeria	Millet	1.12	24.51
Kenya	Cassava	10.76	25.67
Ghana	Maize	1.69	27.65
Ghana	Millet	0.97	30.86
Mali	Cassava	12.12	32.06
Ethiopia	Maize	2.37	33.15
Kenya	Millet	0.65	34.24
Mali	Maize	1.87	34.99
Ghana	Cassava	14.52	37.30
Ethiopia	Millet	1.53	42.96

A Kruskal-Wallis test comparing the distribution of CV% across the three crops returned $H = 0.360$, $p = 0.835$. $H0_3$ is therefore not rejected: there is no statistically significant difference in yield stability among Maize, Millet and Cassava at the national level. This result does not support the initial expectation that Maize would show systematically higher instability than the more drought-tolerant Millet or the more resilient Cassava; instead, the country-to-country range within each crop is comparable to or larger than the range between crops, suggesting that country-level factors are a stronger driver of yield instability than crop identity alone.

The rolling 10-year CV trend (Figure 3) provides a temporal view of this stability question. Millet shows the clearest upward trend in instability across multiple countries, particularly Kenya and Nigeria, both rising sharply through the 2010s into the 2020s, with Ethiopia also showing elevated and rising instability before the series ends. Maize instability shows no consistent directional trend across countries, though Mali Maize shows a pronounced single-decade spike (CV approaching 37%) centred on the 2009–2011 window before declining. Cassava instability is more idiosyncratic, with Nigeria, Kenya and Ghana all showing fluctuating but broadly rising volatility from the mid-2000s onward, while Mali Cassava remains comparatively volatile throughout without a clear trend direction.

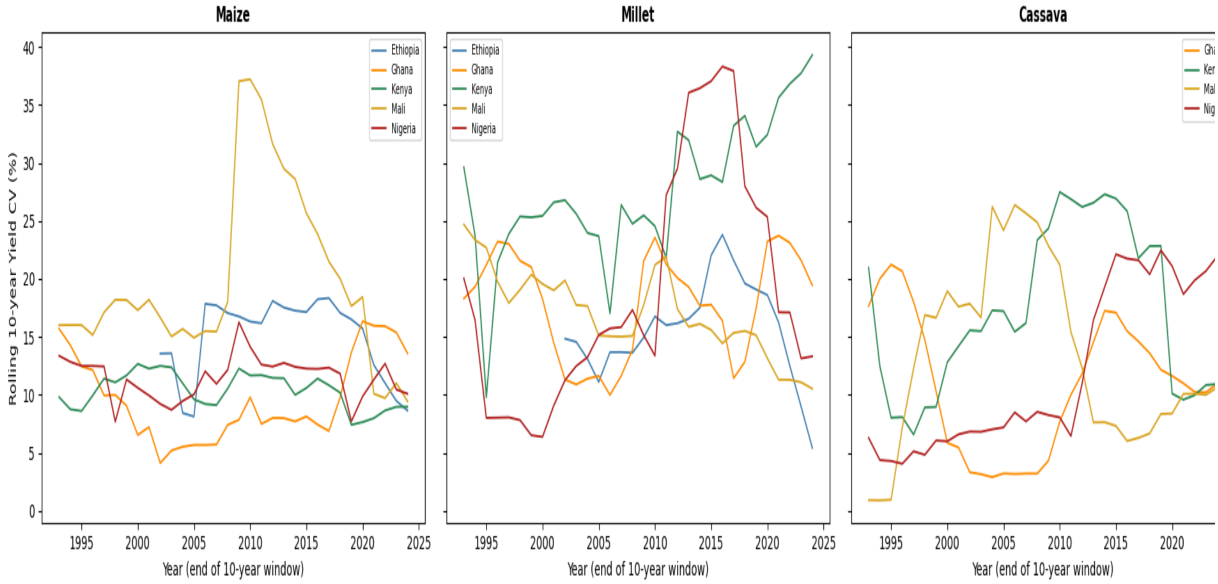

Figure 3. Rolling 10-year yield coefficient of variation (%) by country, faceted by crop.

3.4 Regional Climate Vulnerability (Hypothesis 4)

Rainfall variability (CV%) differs substantially across the five countries (Table 10), ranging from 14.5% in Ghana to 58.1% in Nigeria.

Table 10. Rainfall coefficient of variation (%) by country.

Country	Mean Rainfall (mm/yr)	Rainfall CV (%)
Ghana	1109.2	14.51
Mali	1007.6	15.87
Kenya	1919.8	22.80
Ethiopia	1808.7	23.84
Nigeria	1759.6	58.09

Climate vulnerability, operationalised as the standardised sensitivity of yield to rainfall anomalies pooled across crops within each country, shows a striking pattern (Table 11): Mali (slope = 0.493, $p < 0.001$), Ethiopia (slope = 0.434, $p < 0.001$) and Kenya (slope = 0.422, $p < 0.001$) all show strong, statistically significant positive sensitivity, while Ghana (slope = -0.141, $p = 0.118$) and, notably, Nigeria (slope = -0.124, $p = 0.169$) show weak, non-significant sensitivity despite Nigeria having by far the highest rainfall variability of the five countries.

Table 11. Standardised yield sensitivity to rainfall anomaly, by country (pooled across crops).

Country	Standardised Slope	r	p-value	n
Mali	0.493	0.497	<0.001 ***	123
Ethiopia	0.434	0.438	<0.001 ***	64
Kenya	0.422	0.426	<0.001 ***	123
Ghana	-0.141	-0.142	0.118	123
Nigeria	-0.124	-0.125	0.169	123

*** $p < 0.001$.

A one-way ANOVA testing whether this standardised rainfall-sensitivity differs across countries (computed at the country-crop level) returned $F = 5.512$, $p = 0.016$. $H0_4$ is therefore rejected: there is a statistically significant difference in climate vulnerability among the five countries, supporting $H1_4$.

4. Discussion

The findings of this study tell a story of climate change that is real but uneven, a climate-yield relationship that is statistically detectable but agronomically modest, and a yield stability landscape where country-specific context consistently matters more than crop identity. Four hypotheses were tested with different outcomes.

Hypothesis 1 on climate trends was partially supported. The results clearly show that the climate is changing, but not everywhere in the same way. The economies of Sub-Saharan African countries depend heavily on a system of agricultural production that is weather and climate-sensitive, making them constantly susceptible to shifting temperature and precipitation patterns (Kotir, 2011). However, the spatial heterogeneity of those shifts is striking: Nigeria, Ghana and Ethiopia show statistically significant warming trends over the study period, while Kenya and Mali show no significant warming but instead show statistically significant increases in annual rainfall. This means that climate adaptation in Sub-Saharan Africa cannot be a one-size-fits-all intervention. A policy, seed variety, or farming practice designed around a warming-and-drying scenario may be entirely the wrong prescription for a country whose primary climate trajectory is toward more rainfall rather than less. The five countries in this study alone already require at least two distinct adaptation frameworks, and across the full breadth of the continent the diversity will be far greater still. This finding justifies the kind of country-specific, long-run empirical work this study represents, precisely because regional averages conceal more than they reveal (Wollburg et al., 2024).

Hypothesis 2 was also rejected: climate variables were found to have a statistically significant joint effect on crop yields in all three crops tested, confirming that the link between climate variability and food production in this region is not a theoretical concern but an empirically detectable reality (Omokpariola et al., 2025). However, the magnitude of that relationship tells an equally important story: climate explains at most 13.5% of yield variance in cassava, 7.7% in millet, and only 4.5% in maize. The remaining 86–95% of what drives national yield variation year to year is something other than the three climate variables measured here. Input availability, seed quality, fertiliser access, post-harvest losses, road infrastructure, price incentives, and political stability all shape what a country's farmers manage to produce in any given year, and none of those factors are captured in a rainfall or temperature time series (Omokpariola et al., 2025). The practical implication is sobering: even if it were possible to perfectly stabilise the climate tomorrow, the majority of yield variability in these countries would remain. Climate adaptation is necessary, but it operates on top of a much larger set of structural agricultural challenges that climate policy alone cannot address.

Hypothesis 3 on yield stability was not supported. Yield instability is more about where you are than what you grow. If asked which crop would show the highest yield instability, most agronomists would say maize, with millet and cassava expected to be more stable. The data do not support this expectation. There is no statistically significant difference in yield stability among the three crops at the national level. The range of instability within any single crop across the five countries is comparable to, or larger than, the range between crops: Kenya's maize is among the most stable crop-country combinations in the dataset, while Ethiopia's millet is among the least stable, and that distinction is driven by country, not crop. What this tells us is that yield stability is ultimately an outcome of the whole farming system, not a property of the seed alone.

Hypothesis 4 on regional climate vulnerability was supported: countries differ significantly in their standardised yield sensitivity to rainfall anomalies. The most striking result in this section is the apparent disconnect between Nigeria's rainfall variability and its near-zero, non-significant yield sensitivity to rainfall. On the surface this looks like a contradiction: how can a country whose rainfall is so dramatically variable show no corresponding yield response? The answer lies in what happened in 2020, 2021 and 2022. Nigeria experienced catastrophic flooding events during those years, displacing over 1.4 million people and destroying more than 332,000 hectares of farmland (Wikipedia, 2024; World Weather Attribution, 2022). These floods were real, severe, and agriculturally devastating. But they were geographically concentrated in riverine states in south-central Nigeria, and their losses were absorbed within the national yield average by unaffected production across the rest of Nigeria's vast and geographically dispersed agricultural system. This points to a fundamental limitation of what national-level yield statistics can and cannot tell us about climate impact: they are well-suited instruments for detecting chronic, widespread climate stress, but are structurally limited in their ability to detect acute, spatially concentrated shocks, regardless of their severity. If we rely on national yield data as the primary indicator of climate impact, we will systematically underestimate the severity of concentrated shocks and systematically miss the farmers most acutely affected.

5. Conclusions

This study examined four decades of climate and yield data for three staple crops across five Sub-Saharan African countries, finding that climate change is real and detectable in the region but heterogeneous in its expression rather than a uniform regional shift. Climate variables jointly and significantly influence yields across all three crops, though their modest explanatory power (4.5–13.5%) reveals that structural factors such as input access, infrastructure, and institutional capacity are at least as important as climate in determining national yield outcomes. No significant difference in yield stability was found among the three crops, suggesting that country-level farming system context outweighs crop physiological traits as a driver of yield reliability. Countries differ significantly in climate vulnerability, with the countries experiencing the strongest vulnerability not necessarily being the ones experiencing the most dramatic climate change, but rather the ones where the combination of climate variability and limited structural buffering capacity leaves yield most exposed to what the atmosphere delivers in any given year.

6. References

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LINK TO GITHUB : https://github.com/brightliam20-ops/AVCAD_FInalProject